# HW2 BPE tokenizer report - Diyar Husayyan, 214034621

**Implementation.** All three tokenizers are from-scratch character-up BPE with a mergeable space marker, a one-internal-space (bigram) cap, byte fallback that never emits UNK, pickle-safe state, and length-faithful decode for the fixed NER aligner. Unlike basic/pre-tokenized BPE it allows selected cross-word merges while forbidding 3+-word tokens.

**Speed (encode).** Merges are applied greedily in rank order (tiktoken-style). I replace the naive per-merge rescan ($O(n^2)$) with a min-heap keyed by merge rank over an intrusive doubly-linked list, so each merge re-touches only its two neighbor pairs — an $O(n \cdot \log n)$ loop that is byte-identical to a slow apply-in-order oracle (verified on thousands of dev lines plus edge cases) yet ~15× faster cold. A per-text cache and weighted-chunk incremental training keep both directions fast; output, efficiency, and F1 are unchanged.

**Tokenizer choices.** tok1 (Twitter): a domain-1-heavy 2:1 (Twitter:news) mix at vocab 5500 → dev F1 0.4817, beating a balanced 1:1 mix at vocab 6000 (0.4780) on all three axes at once. Pure-domain-1 scored lower (≤0.4352) and larger vocabs cut F1 — the news data regularizes the Twitter vocabulary while the 2:1 ratio keeps Twitter coverage. (A structured pre-tokenization variant was tried and reverted for lowering F1 — a deliberate negative result.) tok2 (news): vocab 12000, chosen by ranking F1/efficiency/speed jointly — F1 0.9606, faster cold encode than vocab 16000 (0.9557) for ~5% more tokens/char. tok3 (hidden): the balanced 1:1 mix at vocab 6000 — deliberately not domain-1-heavy, so it generalizes rather than overfits to Twitter — plus byte fallback; visible-domain d1=0.4780, d2=0.9621, zero UNK, exact OOV reconstruction.

| model/split | vocab | dev F1 | tok/char | tok/sent | cold tok/s |
|---|---|---|---|---|---|
| tok1 domain_1 | 5500 | 0.4817 | 0.306 | 22.49 | 278600 |
| tok2 domain_2 | 12000 | 0.9606 | 0.241 | 21.77 | 187317 |
| tok3 domain_1 | 6000 | 0.4780 | 0.307 | 22.52 | 267020 |
| tok3 domain_2 | 6000 | 0.9621 | 0.287 | 25.93 | 270166 |

**Training-set bigrams.** Full-file counts of adjacent two-word sequences (the assignment's "bigram"). The tail is heavy 3.9M of tok1's 4.7M distinct pairs occur once so the "least 5" are the alphabetically-first singletons and the "top 5" are dominant function-word pairs.

| tok | train file | top 5 two-word sequences | least 5 two-word sequences |
|---|---|---|---|
| tok1 | data\mix_d1x2_d2all_seed42.txt | in the (30267); of the (25083); for the (18616); to the (18066); on the (17280) | ! !!! (1); ! #Apprentice (1); ! #FB (1); ! #FrenchMcflyTeam (1); ! #Myspace-y (1) |
| tok2 | data\domain_2_train.txt | of the (15662); in the (12929); to the (6643); on the (5665); to be (5548) | !! Shot (1); !!! It’s (1); !&*@#%! Watch (1); !@#$-ing Kidding (1); !T Books (1) |
| tok3 | data\tok3_mix_d1sample_d2all_seed42.txt | in the (21479); of the (20299); to the (12293); for the (11919); on the (11441) | ! #FB (1); ! #MetaKong (1); ! #Myspace-y (1); ! #SMTB (1); ! #familyforce5 (1) |

**Cross-word bigrams (studied tradeoff).** Each tokenizer ships exactly one checker-defined bigram token (comma+quote + the space-marker form of "the"). I also built a reproducible collocation phase over whole alphabetic word pairs: on tok1, K=40 collocations (e.g. I am, How To, Donald Trump) cut tokens/sentence by only 0.08% (22.5193→22.5008) while risking named-entity joins, so the code ships K=0 — one bigram, zero trigrams, zero UNK.

**Hidden domain & model-2.** tok3 generalizes by design: the balanced mix spans both registers, medium vocab 6000 avoids overfitting, and byte fallback gives zero UNK with exact reconstruction, so unseen words degrade to short subwords rather than failing. Expected hidden-domain F1 lies between the visible domains (≈0.29–0.31 tok/char), tracking the closer register. tok2's high news F1 reflects stable formal-text boundaries and repeated entity/title patterns that vocab 12000 compresses without fragmenting labels — beating vocab 16000 on F1 and speed.